

CIV102 Project Team 502 Design Calculations

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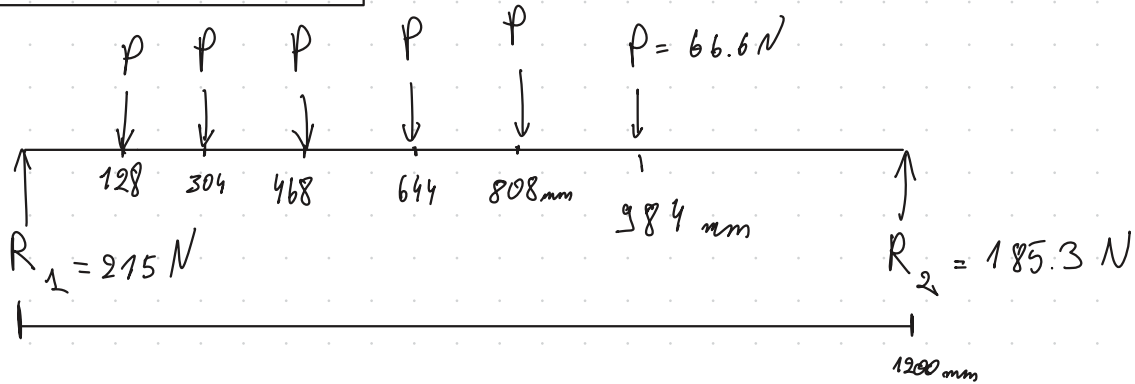
November 23, 2025

Section 1: Design 0 Calculations

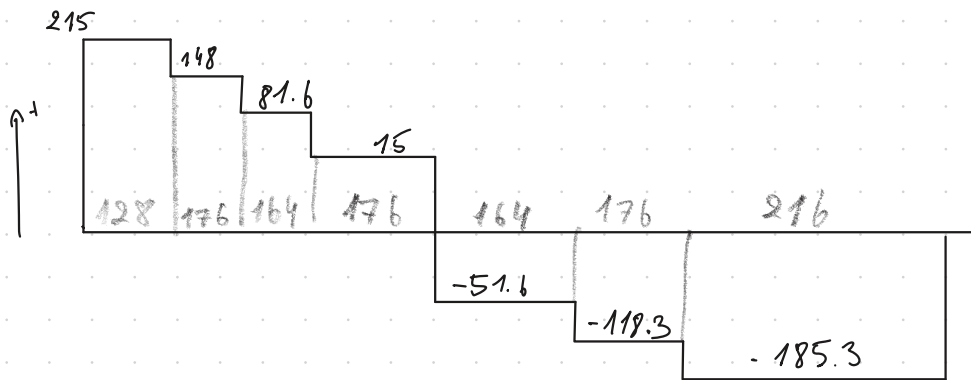
The Following Pages Contain the Hand Calculations for Design 0. The calculations for the centroidal axis and the moment of inertia are not included since they are in deliverable 1.

Design Calculations

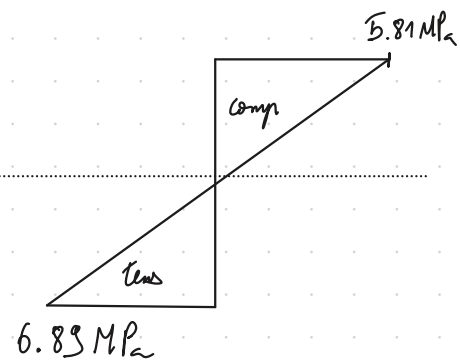
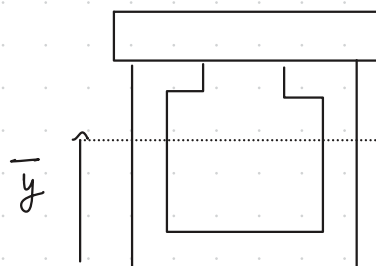
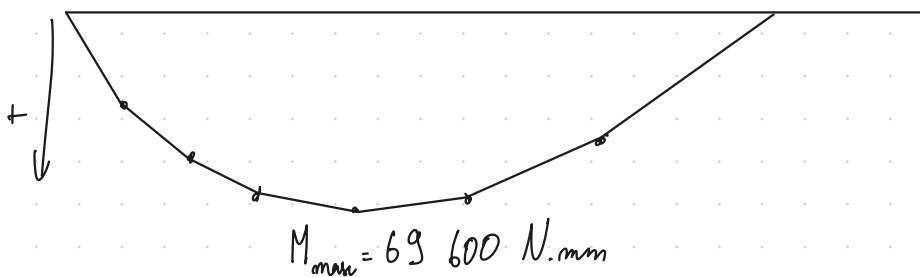
Case max 1



SFD:



BMD:



Flexural stress

$$* \sigma_{c, \max} = 5.81 \text{ MPa}$$

$$FoS = \frac{\sigma_c'}{\sigma_{c, \max}} = \frac{6}{5.81} = 1.033$$

$$* \sigma_{t, \max} = 6.89 \text{ MPa}$$

$$FoS = \frac{\sigma_t'}{\sigma_{t, \max}} = \frac{30}{6.89} = 4.35$$

Shear stress

$$* \tau_{\max} = \frac{V_{\max} \cdot Q_{\max}}{I \cdot b_{\text{centroid}}} = 1.256 \text{ MPa}$$

$$FoS = \frac{\tau_m'}{\tau_{\max}} = \frac{4}{1.256} = 3.18$$

with $b = 254 \text{ mm}$

$$Q_{\max} = \sum A \times d \quad \text{at centroidal axis}$$

$$Q_1 = 80 \times 1.27 \times 40.8 = 4150 \text{ mm}^3$$

$$Q_2 = 2 \times 1.27 \times 40.1 \times 20.1 = 2060 \text{ mm}^3$$

$$Q_{\max} = 6200 \text{ mm}^3$$

$$* \tau_{g, \max} = \frac{V_{\max} \cdot Q_{\text{glue}}}{I \cdot b_{\text{glue}}} = 0.223 \text{ MPa}$$

$$FoS = \frac{\tau_g'}{\tau_{g, \max}} = \frac{2}{0.223} = 8.97$$

$b_{\text{glue}} = 10 \text{ mm}$

$$Q_{\text{at glue}} = A \times d$$

$$= 100 \times 1.27 \times 34.2$$

$$= 4340 \text{ mm}^3$$

Buckling

Case 1

$$\begin{aligned} * \sigma_{crit} &= \frac{4\pi^2 \times E}{12(1-\mu^2)} \times \left(\frac{t}{b}\right)^2 & \text{with } t &= 1.27 \\ & & b &= 80 \\ & & \mu &= 0.2 \\ & & E &= 4000 \\ &= \frac{4\pi^2 \times 4000}{12(1-0.2^2)} \times \left(\frac{1.27}{80}\right)^2 \\ &= 3.45 \text{ MPa} \end{aligned}$$

$$F_o S = \frac{\sigma_{crit}}{\sigma_{c, max}} = \frac{3.45}{5.81} = 0.595$$

Case 2

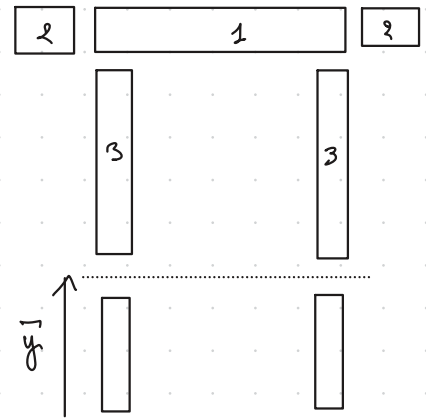
$$\begin{aligned} * \sigma_{crit} &= \frac{0.425\pi^2 E}{12(1-\mu^2)} \times \left(\frac{t}{b}\right)^2 & \text{with } t &= 1.27 \\ & & b &= 10 \\ &= \frac{0.425\pi^2 \times 4000}{12(1-0.2^2)} \times \left(\frac{1.27}{10}\right)^2 \\ &= 23.5 \text{ MPa} \end{aligned}$$

$$F_o S = \frac{\sigma_{crit}}{\sigma_{c, max}} = \frac{23.5}{5.81} = 4.04$$

Case 3

$$\begin{aligned} * \sigma_{crit} &= \frac{6\pi^2 E}{12(1-\mu^2)} \times \left(\frac{t}{b}\right)^2 & t &= 1.27 \\ & & b &= 75 - \bar{y} = 33.6 \\ &= \frac{6\pi^2 \times 4000}{12(1-0.2^2)} \times \left(\frac{1.27}{33.6}\right)^2 \\ &= 29.4 \text{ MPa} \end{aligned}$$

$$F_o S = \frac{\sigma_{crit}}{\sigma_{c, max}} = 5.26$$



$$\begin{aligned} \sigma_{c, max} &= \frac{M y}{I} \\ &= \frac{69600 \times 33.6}{0.418 \times 10^6} \\ &= 5.59 \end{aligned}$$

Case 4

$$\ast \tau_{crit} = \frac{5 \pi^2 E}{12(1-\mu^2)} \left[\left(\frac{t}{a} \right)^2 + \left(\frac{t}{h} \right)^2 \right]$$

with $t = 1.27$
 $a = 400$
 $h = 75$

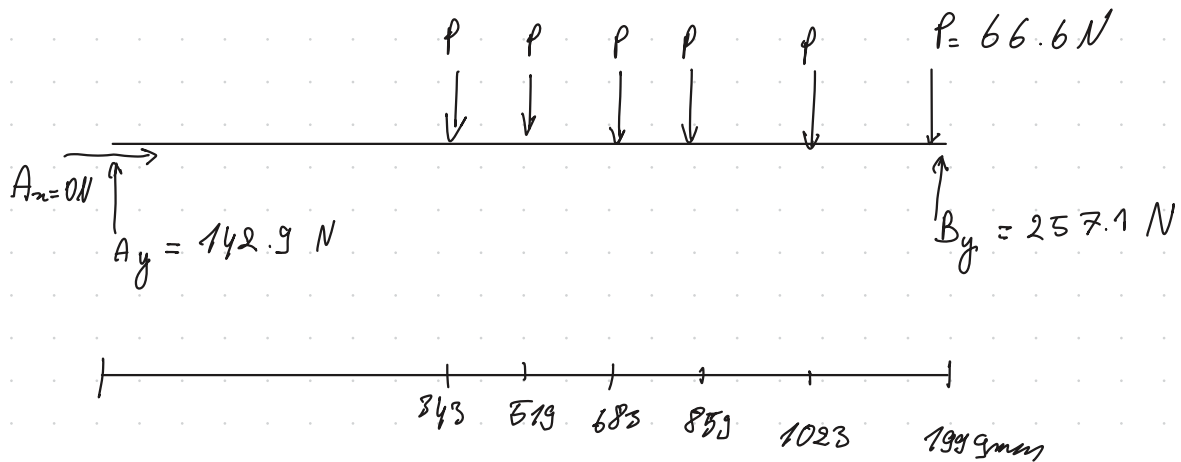
$$= \frac{5 \pi^2 \cdot 4000}{12(1-0.2^2)} \left[\left(\frac{1.27}{400} \right)^2 + \left(\frac{1.27}{75} \right)^2 \right]$$

$$= 5.09$$

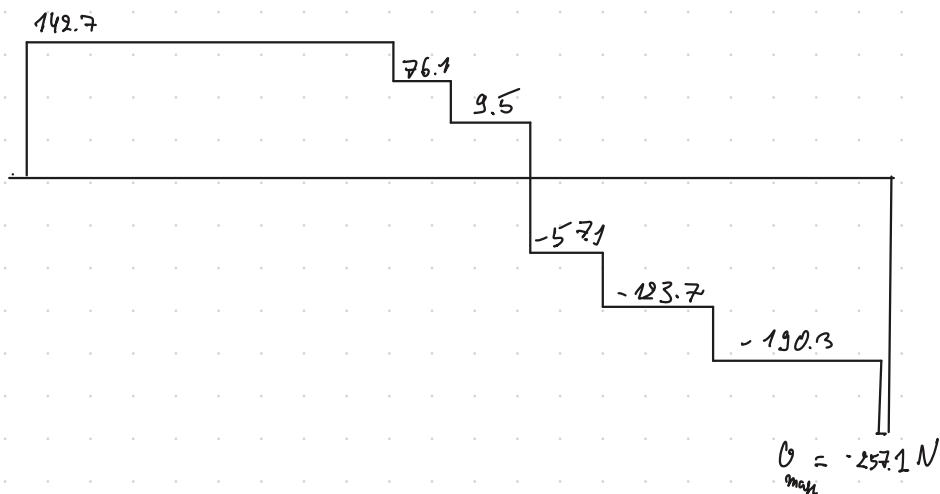
$$FoS = \frac{\tau_{crit}}{\tau_{max}} = \frac{5.09}{1.256} = 4.05$$

Case Max 2

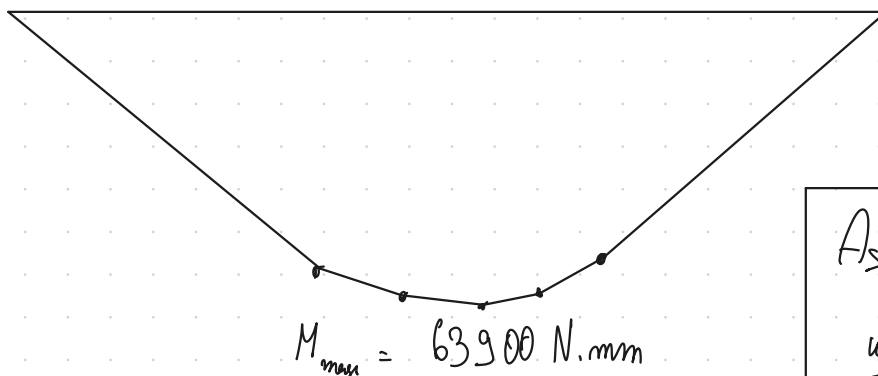
FBD:



SFD:



BMD:



As $M_{max, \text{case 1}} > M_{max, \text{case 2}}$,
we are not checking for flexural stresses and case 1,2 local buckling stresses.

Shear stress

$$* \tau_{max} = \frac{v_{max} \cdot Q_{max}}{I \cdot b_{centroid}}$$

$$= \frac{257.1 \times 6200}{0.418 \times 10^6 \times 2.54}$$

$$= 1.497 \text{ MPa}$$

$$\left\{ \begin{array}{l} \text{with } b = 2.54 \text{ mm} \\ Q_{max} = 6200 \text{ mm}^3 \end{array} \right.$$

$$FoS = \frac{4}{1.497} = 2.67$$

$$* \tau_{g,max} = \frac{v_{max} \cdot Q_{glue}}{I \cdot b_{glue}}$$

$$= \frac{257.1 \times 4340}{0.418 \times 10^6 \times 10}$$

$$= 0.2668 \text{ MPa}$$

$$\left\{ \begin{array}{l} b_{glue} = 10 \text{ mm} \\ Q_{glue} = 4340 \text{ mm}^3 \end{array} \right.$$

$$FoS = \frac{2}{0.267} = 7.49$$

Local Buckling

Case 4

$$* \tau_{crit} = 5.09$$

$$FoS = \frac{\tau_{crit}}{\tau_{max}} = \frac{5.09}{1.497} = 3.40$$

Conclusion for Design 0:

The failure mode is case 1 local buckling with a FoS of 0.595.

Code Calculations

The following results were obtained using the code displayed in [section 3](#) for design zero under load case 1. Only the raw output is displayed as per assignment expectation, thus, refer to the design report for values reported with correct significant figures.

Code Output	Value
{'Centroidal Axis from Bottom (mm)'} }	41.431
{'Second Moment of Inertia (mm ⁴)' }	4.1835e+05
{'Maximum Shear' }	257
{'Maximum Bending Moment' }	69345
{'FOS Against Flexural Compression' }	1.039
{'FOS Against Flexural Tension' }	4.3684
{'FOS Against Centroidal Shear' }	2.6704
{'FOS Against Glue Shear' }	7.4948
{'FOS Against Buckling Case 1' }	0.59821
{'FOS Against Buckling Case 2' }	4.0678
{'FOS Against Buckling Case 3' }	5.2891
{'FOS Against Shear Buckling' }	3.3954
{'Percentage of Material Used' }	55.616
{'Maximum Load (N)' }	239.28
{'Efficiency Index' }	0.041014

Figure 1: A screenshot of a summary of all significant and intermediate code outputs for design zero under load case 1.

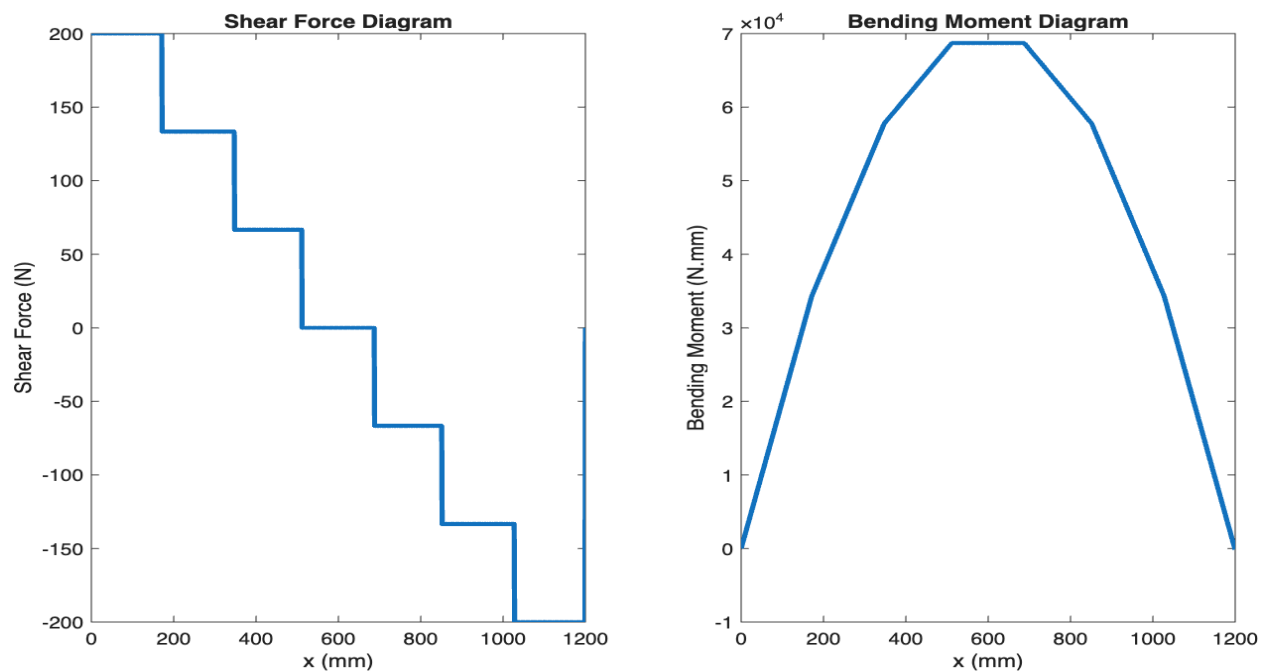


Figure 2: Code output of the shear force diagram and bending moment diagram for design zero under load case 1 with the train in the middle of the bridge.

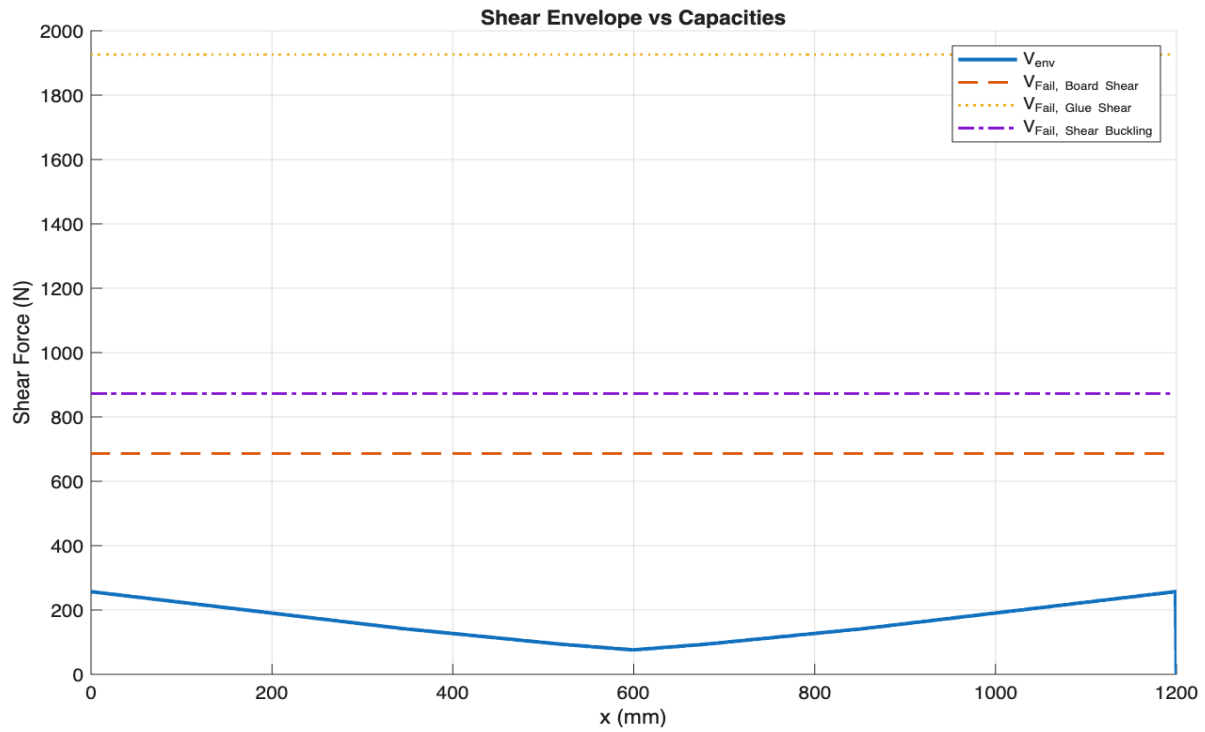


Figure 3: A graph displaying the shear force capacities and the shear force envelope (SFE) for all load case 1 train locations on the design zero bridge. No failure is shown here.

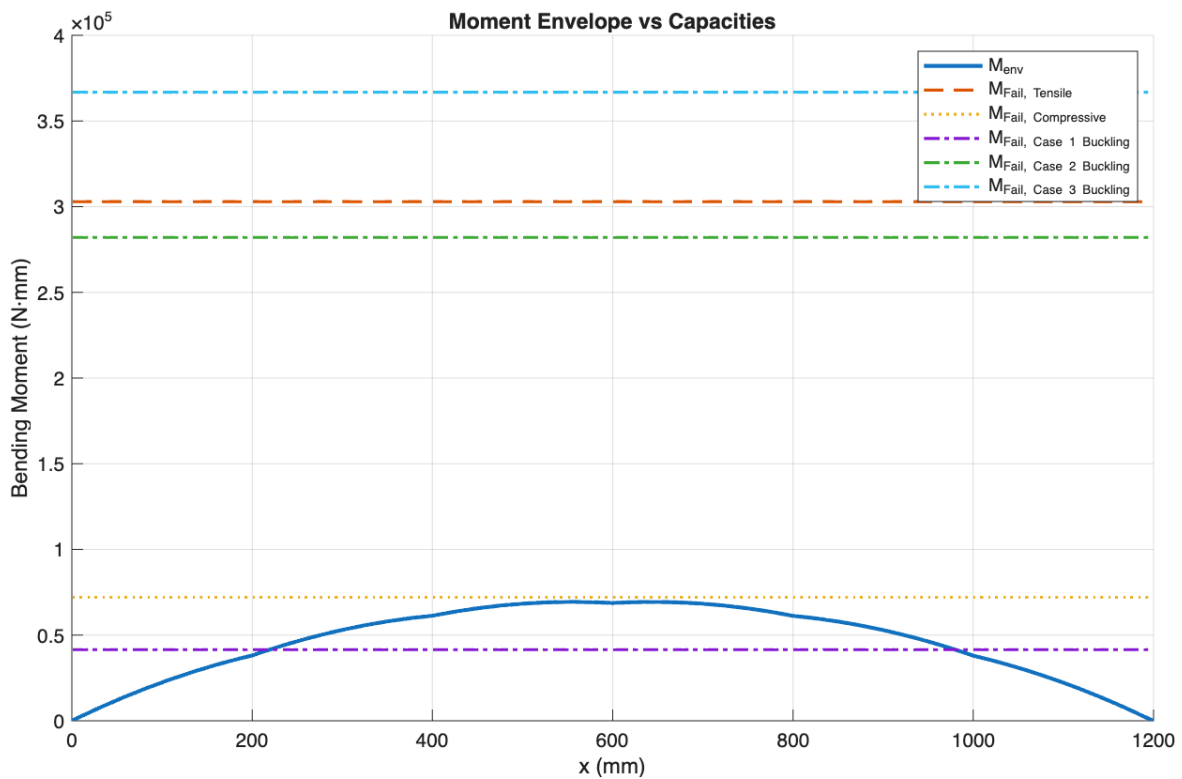


Figure 4: A graph displaying the bending moment capacities and the bending moment envelope (BME) for all train locations on the bridge. As depicted, the bridge fails due to case 1 buckling.

Section 2: Final Design Calculations (Load Case 2)

The following results were obtained using the code in [section 3](#) for the final design under load case 2. Only the raw output is displayed as per assignment expectation, thus, refer to the design report for values reported with correct significant figures.

Code Output	Value
{'Centroidal Axis from Bottom (mm)'} }	99.005
{'Second Moment of Inertia (mm ⁴)' }	1.2889e+06
{'Maximum Shear' }	303.73
{'Maximum Bending Moment' }	77964
{'FOS Against Flexural Compression' }	2.4196
{'FOS Against Flexural Tension' }	5.0095
{'FOS Against Centroidal Shear' }	3.4635
{'FOS Against Glue Shear' }	8.4116
{'FOS Against Buckling Case 1' }	7.2785
{'FOS Against Buckling Case 2' }	16.842
{'FOS Against Buckling Case 3' }	9.6414
{'FOS Against Shear Buckling' }	2.33
{'Percentage of Material Used' }	86.302
{'Maximum Load (N)' }	1053.2
{'Efficiency Index' }	0.11633

Figure 5: A screenshot of a summary of all significant and intermediate code outputs for the final design under load case 2.

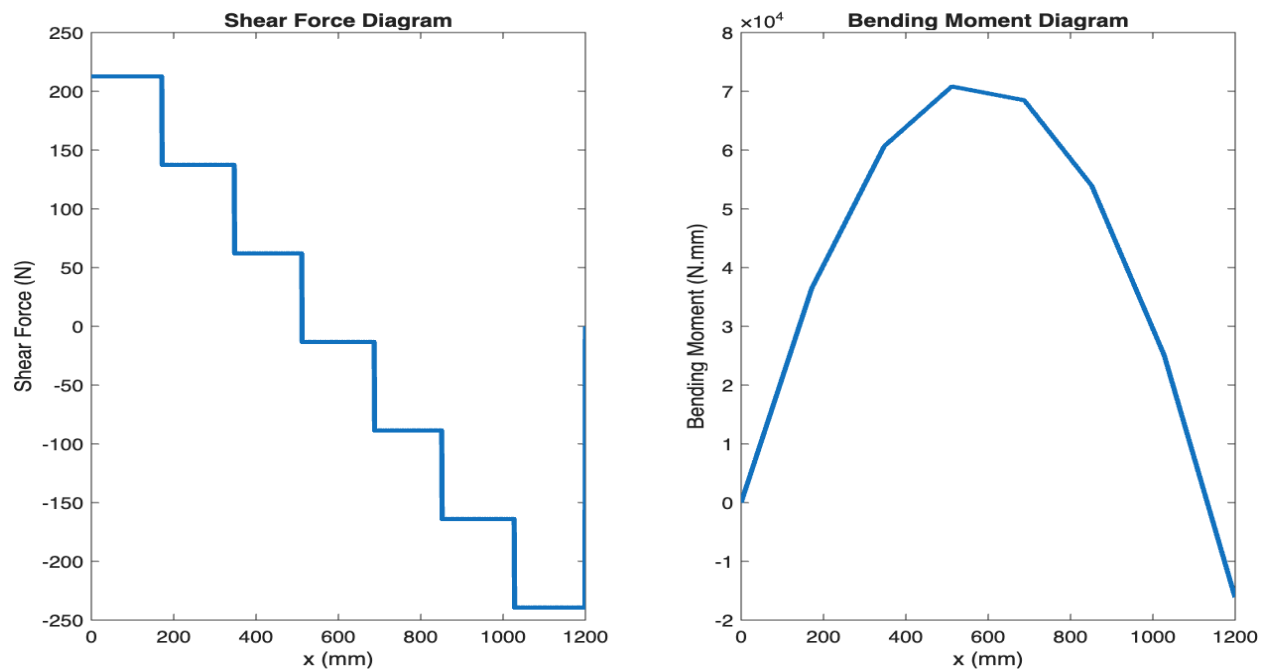


Figure 6: Code output of the shear force diagram and bending moment diagram for the final underload case 2 with the train in the middle of the bridge.

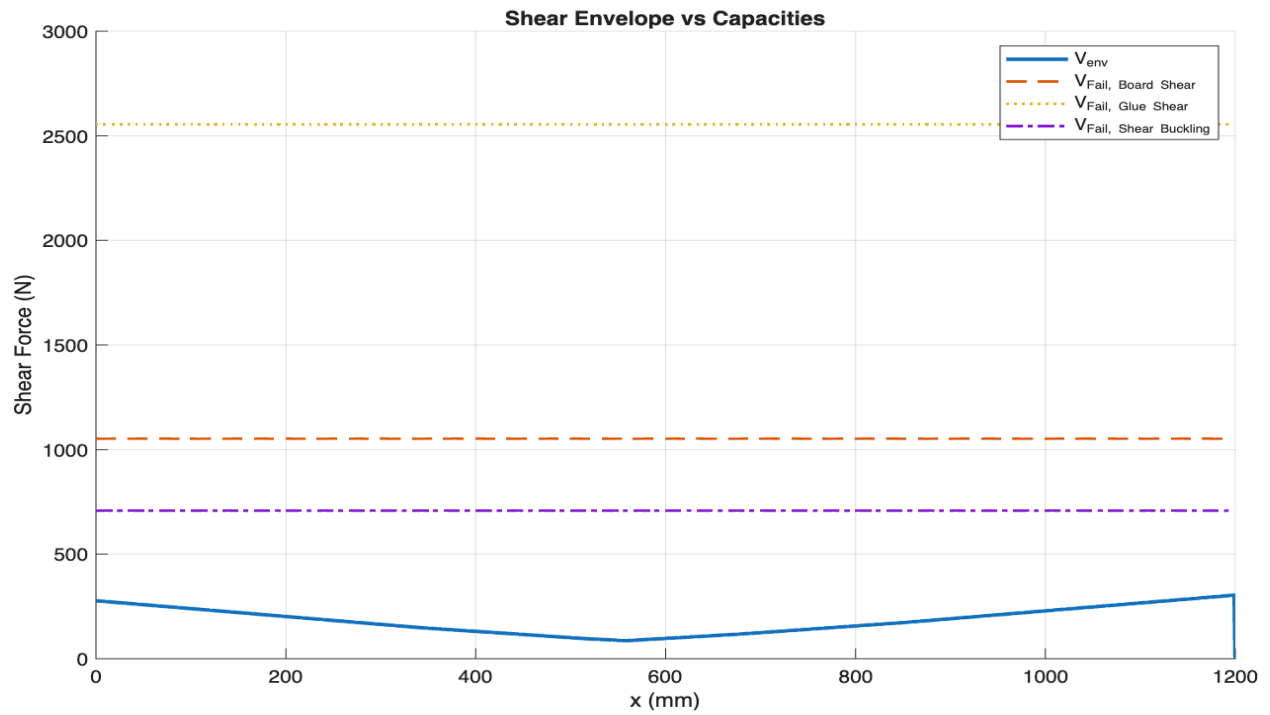


Figure 7: A graph displaying the shear force capacities and the shear force envelope (SFE) for all load case 2 train locations on the final bridge design. No failure is shown here.

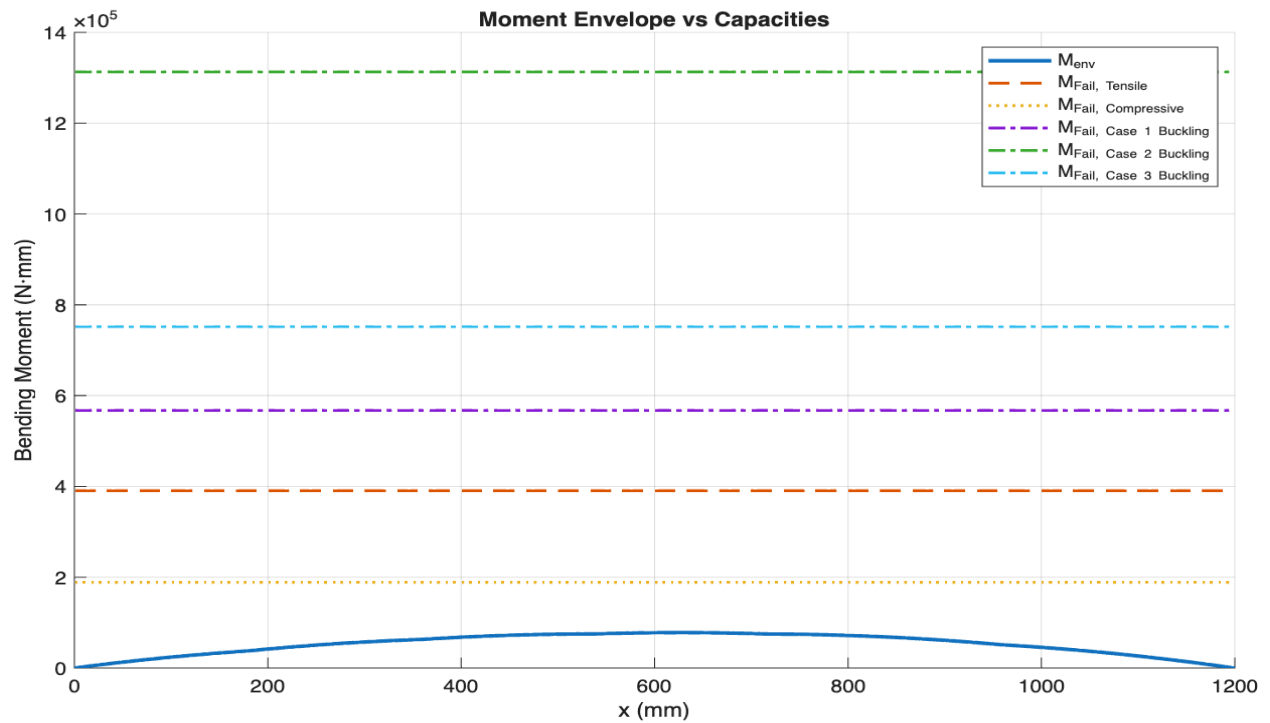


Figure 8: A graph displaying the bending moment capacities and the bending moment envelope (BME) for all load case 2 train locations on the final bridge design. No failure is shown here.

Section 3: The Code

The code below is a MatLab Live Script with examples of outputs for design 0.

*****DISCLAIMER: If you run the code, it may take a few minutes since it generates a graph for each point along the track.**

Other Notes:

- *For all iterations (and design zero), only the tab width was used for any calculations involving glue shear since we knew that in practice, the thickness of the matboard would not count due to the shape of bending. This practice also provided safer calculations.*

Bridge Design Code (Team 502)

***DISCLAIMER: This code takes a very long time to run. Expect it to take MINUTES. It does its job well, but it takes time to do it.

Defining the Cross Section:

```
%%% Create a Figure with Appropriate Bounds (This is based on the fact that the
assignment doesn't allow our bridges to be over 200mm high) %%%
dimensions = [-200 200 -200 200];
figure(Name="Cross-Section");
axes;
axis(dimensions);
title('Design x Cross Section');
xlabel('Length (mm)');
ylabel('Height (mm)');

%%% Define the Properties of the CrossSection %%%
thickness = 1.27;
basewidth = 80; %%% Must be Able to Hold The Trains
basethickness = thickness;
topflangethickness1 = thickness;
topflangethickness2 = 0;
totalbeamheight = 76.27;
topwidth = 100;
topwidthsecondlayer = 0;
tabwidth = 5;
%%% ----- %%%
%%%DEFINE THE DIAPHRAGMS
DiaphragmSpacing = 400; %%% Max Diaphragm Spacing
```

```

NumberDiaphragms = 6;
%%% ----- %%%

%%% DEFINE THE LOAD CASE:
% %CASE 1 (comment out if not used)
P = 400
P_train = [1 1 1 1 1 1] * P/6;

% %CASE 2 (comment out if not used)
% P_train = [67.5 67.5 67.5 67.5 91 91];
% P = 452

%%% ----- %%%

%% Now compute some important values based on the parameters...:

totaltoplayerthickness = topflangethickness1 + topflangethickness2;
heightwithouttopflange = totalbeamheight - topflangethickness1 -
topflangethickness2;
heightrectweb = totalbeamheight - topflangethickness1 - topflangethickness2 -
basethickness;

BaseCentroid = basethickness/2;
WebCentroid = basethickness + heightrectweb/2;

totalheight = basethickness + heightrectweb + totaltoplayerthickness;

AreaDiaphragm = (heightrectweb-thickness)*(basewidth-2*thickness) + (basewidth
- 2*tabwidth - 2*thickness)*thickness;
VolumeDiaphragms = AreaDiaphragm * thickness * NumberDiaphragms;

%%% Initialize the Desired Cross Section:
outeredgehpos = basethickness;
rightouteredgexpos = basewidth/2 - thickness;
leftouteredgexpos = -rightouteredgexpos;
toppos = basethickness + heightrectweb;
toppossecondlayer = toppos + topflangethickness1;

%%% Define the Geometry %%%

%FOR DEFINING RECTANGLES:
% rectangle('Position', [xposition yposition length height]

```

```

%BASE STRUCTURE GEOMETRY
baserectangle = [-basewidth/2 0 basewidth basethickness];

webrectangleright = [rightouteredgexpos outeredgehpos thickness heightrectweb];

webrectangleleft = [leftouteredgexpos-thickness outeredgehpos thickness
heightrectweb];

toprectangle1 = [-topwidth/2 toppos topwidth topflangethickness1];

toprectangle2 = [-topwidthsecondlayer/2 toppossecondlayer topwidthsecondlayer
topflangethickness2];

rectangle('Position', baserectangle);
rectangle('Position', webrectangleright);
rectangle('Position', webrectangleleft);
rectangle('Position', toprectangle1);
rectangle('Position', toprectangle2);

%GLUETABS

xposrighttab = basewidth/2 - thickness - tabwidth;
xposlefttab = -basewidth/2 + thickness;

height = heightrectweb + basethickness - thickness;

% Define the positions for the glue tabs
tab3 = [xposrighttab height tabwidth thickness];
tab4 = [xposlefttab height tabwidth thickness];

%% Place them
rectangle('Position', tab3);
rectangle("Position", tab4);

%% IF GLUETABS NEEDED ON THE BOTTOM

% tab1 = [xposright thickness tabwidth thickness];
% tab2 = [xposleft thickness tabwidth thickness];
%
% rectangle("Position", tab1);
% rectangle("Position", tab2);

```

Getting the YBar Value

```
ABase = basewidth*basethickness;  
DBase = basethickness/2;  
  
AWeb = thickness * heightrectweb;  
DWeb = basethickness + heightrectweb/2;  
  
ATop = topflangethickness1 * topwidth;  
DTop = basethickness + heightrectweb + topflangethickness1/2;  
  
ATop2 = topflangethickness2 * topwidthsecondlayer;  
DTop2 = basethickness + heightrectweb + topflangethickness1 +  
topflangethickness2/2;  
  
AGlueTab = thickness*tabwidth;  
DTopTabs = basethickness + heightrectweb - thickness/2;  
  
yBar = (ABase*DBase + 2*AWeb*DWeb + ATop*DTop + ATop2*DTop2 +  
2*AGlueTab*DTopTabs)/(ABase + 2*AWeb + ATop + 2*AGlueTab + ATop2)
```

Getting the Second Moment of Intertia

```
IBottom = (basewidth * (basethickness^3))/12;  
dBottom = abs(yBar - DBase);  
  
IWeb = (thickness * (heightrectweb^3))/12;  
dWeb = abs(yBar - DWeb);  
  
ITop = (topwidth*(topflangethickness1^3))/12;  
dTop = abs(yBar - DTop);  
  
ITop2 = (topwidthsecondlayer*(topflangethickness2^3))/12;  
dTop2 = abs(yBar - DTop2);  
  
ITab = (tabwidth*(thickness^3))/12;  
dTopTab = abs(yBar - DTopTabs);
```

%%% Parallel Axis Theorem.


```
I = IBottom + ITop + ITop2 + 2*IWeb + 2*ITab + ABase*(dBottom^2) +  
ATop2*(dTop2^2) + ATop*(dTop^2) + 2*AWeb*(dWeb^2) + 2*AGlueTab*(dTopTab^2)
```

Create an SFD and BMD for the Middle Case

```
figure(Name="SFD and BMD (Middle Position)");  
  
% Initialize Parameters (BASE CASE)  
  
L = 1200; % Length of bridge  
n = 1200; % Discretize into 1 mm seg.  
x = linspace(0, L, n+1); % x-axis  
  
x_train = [172 348 512 688 852 1028];  
  
totalLoad = 0;  
  
%%% Initialize The max.  
ShearMax = 0;  
MomentMax = 0;  
  
%%%Without Rolling, Just Assume in Middle First:  
  
%%% FIND THE REACTION FORCES  
IndMoments = x_train.*P_train;  
RightReaction = sum(IndMoments) / L;  
  
    for i = 1:6  
        if x_train(i) > 0  
            totalLoad = totalLoad + P_train(i);  
        end  
    end  
  
LeftReaction = totalLoad - RightReaction;  
  
SFT1 = LeftReaction - P/6;  
SFT2 = SFT1 - P/6;  
SFT3 = SFT2 - P/6;  
SFT4 = SFT3 - P/6;  
SFT5 = SFT4 - P/6;  
SFT6 = SFT5 - P/6;
```

```

SFR = SFT6 + RightReaction;
SFS = [SFT1 SFT2 SFT3 SFT4 SFT5 SFT6 SFR];

shearforces = [LeftReaction SFS];
shearforceloc = [0 x_train L];

%%% Now Plot Continuous to Look Like a Real SFD

SFD = x.*0;

for i = 0:1200
    if ismember(i, shearforceloc)
        theindex = find(shearforceloc==i);
        SFD(i+1) = shearforces(theindex);
    else
        SFD(i+1) = SFD(i); % Maintain the previous shear force value
    end
end

%%% Pretty much just an integral function but for wonky graphs.

BMD = cumtrapz(x, SFD);

subplot(1,2,1)
plot(x, SFD, LineWidth=2.5)
title("Shear Force Diagram")
ylabel("Shear Force (N)")
xlabel("x (mm)")

subplot(1,2,2)
plot(x, BMD, LineWidth=2.5)
title("Bending Moment Diagram")
ylabel("Bending Moment (N.mm)")
xlabel("x (mm)")

```

Getting the Envelope

```

x_train = [-856 -680 -516 -340 -176 0]; % Train Load Locations

ShearMax = 0;
ShearMaxLoc = 0;

MomentMax = 0;
MomentMaxLoc = 0;

```

```

IndMoments = zeros(1,6);

%graph increments:
graphinc = 1;

figure();

% Envelopes (Add The Maximum Values at each Location)
Venv = zeros(size(x));
Menv = zeros(size(x));

for i = 1:(L + 856) %% So it can fully exit the bridge
    %% FIND THE REACTION FORCES
    totalLoad = 0;

    for m = 1:6
        if x_train(m) > 0 && x_train(m) < L %% Weird matlab syntax but okay
lol.
            totalLoad = totalLoad + P_train(m);
        end
    end

    for j = 1:6
        if x_train(j) > 0 && x_train(j) < L
            IndMoments(j) = x_train(j)*P_train(j);
        else
            IndMoments(j) = 0;
        end
    end

    RightReaction = sum(IndMoments) / L;
    LeftReaction = totalLoad - RightReaction;

    %% Now Construct the SFD:
    %%ShearForce Vectors:

    shearforceloc = zeros(1, 8);
    SFS = zeros(1, 8);

```

```

shearindex = 1;
shearforceloc(shearindex) = 0;
SFS(shearindex) = LeftReaction;

for f = 1:6
    if x_train(f) > 0 && x_train(f) < L
        shearindex = shearindex + 1;
        shearforceloc(shearindex) = x_train(f);
        SFS(shearindex) = SFS(shearindex-1) - P_train(f);
    end
end

shearindex = shearindex + 1;
SFS(shearindex) = 0;
shearforceloc(shearindex) = L;

% TO GET RID OF THE MULTIPLE ZEROS THAT MAY BE PRESENT FOR THE NEXT
% LOOP:
validLocs = shearforceloc(1:shearindex);
validSFS = SFS(1:shearindex);

SFDi = x .* 0;

for n = 0:L
    if ismember(n, validLocs)
        theindex = find(validLocs==n, 1);
        SFDi(n+1) = validSFS(theindex);
    else
        SFDi(n+1) = SFDi(n); % Maintain the previous shear force value
    end
end

BMDi = cumtrapz(x, SFDi); % integrate with respect to x

if max(abs(SFDi)) > ShearMax
    ShearMax = max(abs(SFDi));
end

if max(abs(BMDi)) > MomentMax
    MomentMax = max(abs(BMDi));
end

```

```

    x_train = x_train + 1;
    if mod(i, graphinc) == 0
        subplot(1, 2, 1)
        plot(x, SFDi);
        hold on
        subplot(1, 2, 2)
        plot(x, BMDi);
        hold on
    end

    Venv = max(Venv, abs(SFDi));
    Menv = max(Menv, abs(BMDi));

end
disp(ShearMax)
disp(MomentMax)

```

Failure Methods (Initialize First):

```

TensileStrength = 30;
CompressiveStrength = 6;
ShearStrength = 4;
E = 4000;
Poisson = 0.2;
GlueStrength = 2;

yTop = totalheight - yBar;
OurBridgeLength = 1260;

V = ShearMax;

TopDoubleBoundedWidth = basewidth;
ExtendedFlangeLayer1 = (topwidth - basewidth)/2;
ExtendedFlangeLayer2 = (topwidthsecondlayer - basewidth)/2;

TopExtendedFlangeWidth = max(ExtendedFlangeLayer1, ExtendedFlangeLayer2);

if ExtendedFlangeLayer1 == ExtendedFlangeLayer2
    ExtendedFlangeThickness = topflangethickness1 + topflangethickness2;
else
    ExtendedFlangeThickness = max(topflangethickness2, topflangethickness1);
end

```

Flexural Stresses

```
FlexuralCompression = MomentMax*yTop/I;  
FlexuralTension = MomentMax*yBar/I;  
  
FOSFlexComp = CompressiveStrength/FlexuralCompression  
FOSFlexTens = TensileStrength/FlexuralTension
```

%%% Now For the Envelope:

```
FlexuralCompressionEnv = Menv * yTop / I;  
FlexuralTensionEnv = Menv * yBar / I;
```

Shear Stresses

Get Q First:

```
%%% We Break into two parts: The Base, and the Portion of the Web.  
%%% The Base is the Easy Part  
  
HeightPartialWebCent = yBar - basethickness;  
APartialWebCent = 2*thickness*HeightPartialWebCent;  
dPartialWebCent = yBar - (HeightPartialWebCent/2 + basethickness);  
  
QBaseCent = ABase*dBottom;  
QPartialWebCent = APartialWebCent*dPartialWebCent;  
  
CentroidalQ = QBaseCent + QPartialWebCent;  
  
%%% Now for the Glue Shear Q (Go from top down):  
  
dGlue = totalheight - topflangethickness2 - topflangethickness1/2 - yBar;  
  
dGlue2 = totalheight - topflangethickness2/2 - yBar;  
  
QGlue = ATop*dGlue + ATop2*dGlue2;
```

Now Evaluate the Shear Stresses Using Formula:

```
MaximumShear = (ShearMax*CentroidalQ)/(I*2*thickness);
GlueShear = (ShearMax*QGlue)/(I*2*(tabwidth));

FOSShearMax = ShearStrength/MaximumShear
FOSGlueShear = GlueStrength/GlueShear

%%% Envelopes:

MaximumShearEnv = (Venv * CentroidalQ) ./ (I * 2*thickness);
GlueShearEnv = (Venv * QGlue) ./ (I * 2*(tabwidth));
```

Buckling Modes

```
%BothRestrained (Things Between Webs) :

CritAllRestrainedTop =
(4*(pi^2)*E)*(((topflangethickness1+topflangethickness2)/TopDoubleBoundedWidth)^
2)/(12*(1-Poisson^2));

FOSCritAllRestrainedTop = CritAllRestrainedTop/FlexuralCompression

%Case2 (Things that are hanging off). :

Crit1RestrainedTop =
(0.425*(pi^2)*E)*((ExtendedFlangeThickness/TopExtendedFlangeWidth)^2)/(12*(1-
Poisson^2));

FOSCrit1RestrainedTop = Crit1RestrainedTop/FlexuralCompression

%Case3 (The Walls From the Centroidal Axis Up):
CritWebs = (6*(pi^2)*E)*((thickness/(basethickness+heightrectweb-
yBar))^2)/(12*(1-Poisson^2));
```

```
Case3Compression = MomentMax * (heightwithouttopflange-yBar)/I;
FOSCritWebs = CritWebs/Case3Compression
```

```
Case3CompressionEnv = Menv * (heightwithouttopflange-yBar)/I;
```

```
%Case 4 (Shear Buckling of the Webs):
```

```
ShearCritWebs = (5*(pi^2)*E) / (12*(1-Poisson^2)) * ( (thickness/(totalheight-  
topflangethickness1-topflangethickness2))^2 + (thickness/DiaphragmSpacing)^2 );
```

```
FOSShearBuckling = ShearCritWebs/MaximumShear
```

FACTOR OF SAFETY ENVELOPES/CAPACITIES FOR PRES (PRESENTATION)

```
FOSFlexCompPres = CompressiveStrength ./ abs(FlexuralCompressionEnv);
```

```
FOSFlexTensPres = TensileStrength ./ abs(FlexuralTensionEnv);
```

```
FOSShearMaxPres = ShearStrength ./ abs(MaximumShearEnv);
```

```
FOSGlueShearPres = GlueStrength ./ abs(GlueShearEnv);
```

```
FOSCritAllRestrainedTopPres = CritAllRestrainedTop ./  
abs(FlexuralCompressionEnv);
```

```
FOSCrit1RestrainedTopPres = Crit1RestrainedTop ./ abs(FlexuralCompressionEnv);
```

```
FOSCritWebsPres = CritWebs ./ Case3CompressionEnv;
```

```
FOSShearBucklingPres = ShearCritWebs ./ abs(MaximumShearEnv);
```

```
% Bending moment capacities
```

```
Mf_tens = FOSFlexTensPres .* Menv;
```

```
Mf_comp = FOSFlexCompPres .* Menv;
```

```
Mf_buck1 = FOSCritAllRestrainedTopPres .* Menv;
```

```
Mf_buck2 = FOSCrit1RestrainedTopPres .* Menv;
```

```
Mf_buck3 = FOSCritWebsPres .* Menv;
```

```
% Shear capacities
```

```
Vf_shear = FOSShearMaxPres .* Venv;
```

```
Vf_glue = FOSGlueShearPres .* Venv;
```

```
Vf_buckV = FOSShearBucklingPres .* Venv;
```


Now Efficiency and Presentations

```
AvailableVolume = 813*1016*thickness;
UsedVolume = (ABase + 2*AWeb + 2*AGlueTab + ATop + ATop2) * OurBridgeLength +
VolumeDiaphragms;
PercentUsed = UsedVolume/AvailableVolume * 100

FactorsOfSafety = [FOSFlexComp FOSFlexTens FOSShearMax FOSGlueShear
FOSCritAllRestrainedTop FOSCRit1RestrainedTop FOSCritWebs FOSShearBuckling];

MaximumPossibleLoad = min(FactorsOfSafety) * P

%%% Get Max Load from Above...

Efficiency = MaximumPossibleLoad/UsedVolume * 100

% Shear capacities
figure;
hold on; grid on;
plot(x, Venv, 'LineWidth', 2);
plot(x, Vf_shear, '--', 'LineWidth', 1.5);
plot(x, Vf_glue, ':', 'LineWidth', 1.5);
plot(x, Vf_buckV, '-.', 'LineWidth', 1.5);
xlabel('x (mm)');
ylabel('Shear Force (N)');
title('Shear Envelope vs Capacities');
legend('V_{env}', 'V_{Fail, Board Shear}', 'V_{Fail, Glue Shear}', 'V_{Fail,
Shear Buckling}');
hold off

% Bending capacities
figure;
hold on; grid on;
plot(x, Menv, 'LineWidth', 2);
plot(x, Mf_tens, '--', 'LineWidth', 1.5);
plot(x, Mf_comp, ':', 'LineWidth', 1.5);
plot(x, Mf_buck1, '-.', 'LineWidth', 1.5);
plot(x, Mf_buck2, '-.', 'LineWidth', 1.5);
plot(x, Mf_buck3, '-.', 'LineWidth', 1.5);
xlabel('x (mm)');
```

```

ylabel('Bending Moment (N·mm)');
title('Moment Envelope vs Capacities');
legend('M_{env}', 'M_{Fail, Tensile}', 'M_{Fail, Compressive}', ...
       'M_{Fail, Case 1 Buckling}', 'M_{Fail, Case 2 Buckling}', 'M_{Fail, Case 3 Buckling}');

%I build a table for readability of the factors of safety:

FailureModes = {'Centroidal Axis from Bottom (mm)'; 'Second Moment of Inertia (mm^4)'; 'Maximum Shear'; 'Maximum Bending Moment'; 'FOS Against Flexural Compression'; 'FOS Against Flexural Tension'; 'FOS Against Centroidal Shear'; 'FOS Against Glue Shear'; 'FOS Against Buckling Case 1'; 'FOS Against Buckling Case 2'; 'FOS Against Buckling Case 3'; 'FOS Against Shear Buckling'; 'Percentage of Material Used'; 'Maximum Load (N)'; 'Efficiency Index'};

FOS_values = [yBar; I; ShearMax; MomentMax; FOSFlexComp; FOSFlexTens; FOSShearMax; FOSGlueShear; FOSCritAllRestrainedTop; FOSCrit1RestrainedTop; FOSCritWebs; FOSShearBuckling; PercentUsed; MaximumPossibleLoad; Efficiency];

FOS_Table = table(FailureModes, FOS_values, 'VariableNames', {'Code Output', 'Value'});

disp(FOS_Table)

```